

Statistical & Numerical Methods HW4

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Technion

(Date: July 10, 2026)

1 Estimators

We know

$$\mu = 5, \quad \sigma^2 = 0.05 \quad (1)$$

1.1 length estimator

The estimator is just

$$\hat{\mu} = \bar{x} = 5.512 \quad (2)$$

1.2 variance estimator

using the formula for an estimator with an unknown μ

$$\sigma^2 = \frac{1}{N-1} \sum_i (x_i - \bar{x})^2 = 0.052 \quad (3)$$

1.3 CL95

We'll assume the distribution of lengths is a normal distribution, so the standard deviation is

$$\sigma' = \frac{\sigma}{\sqrt{N}} = \frac{\sqrt{0.052}}{\sqrt{10}} = 0.072 \quad (4)$$

so

$$\mu_+ = 1.96\sigma' + \mu = 5.653, \quad \mu_- = 5.371 \quad (5)$$

1.4 Brown screws

directly the expected p is

$$p = \frac{b}{N} = 0.6 \quad (6)$$

and for the variance, marking the number of brown screws as b ,

$$\sigma_p^2 = \left(\frac{\partial p}{\partial b} \right)^2 \sigma_b^2 = \frac{\sigma_b^2}{N^2} \quad (7)$$

the number of brown screws pulled is binomial so

$$\sigma_p^2 = \frac{Np(1-p)}{N^2} = \frac{p(1-p)}{N} = \frac{0.6 \cdot 0.4}{100} = 0.0024 \quad (8)$$

so

$$\sigma_p = 0.049 \quad (9)$$

so the CL95 range is

$$\mu_- = p - 1.96\sigma_p = 0.504, \quad \mu_+ = 0.696 \quad (10)$$

2 Radioactive Decay

2.1 5 minutes

The decay rate is known so the different minutes are independent meaning

$$N_5 = 5N_1 \quad (11)$$

but since N_1 is distribution poisson,

$$N_1 = \lambda \pm \sqrt{\lambda} \quad (12)$$

so

$$N_5 = 5\lambda \pm \sqrt{5\lambda} = 500 \pm 10\sqrt{5} \quad (13)$$

2.2 Unknown material

Since we assume the decay is poisson this tells us that

$$\lambda = 100 \quad (14)$$

which implies

$$\sigma = \sqrt{\lambda} = 10 \quad (15)$$

which is a variance whose source is the measurement. We add this to the variance from the last part which is inherent to the decay and get

$$N = 500 \pm 10\sqrt{5} \pm 50 \quad (16)$$

2.3 composite material

Since we measured the number of decays of both materials together and they are independent we measured

$$N_i \sim Pois(\lambda_1 + \lambda_2) \quad (17)$$

so this is not enough to measure each separately. Any attempt to create an equation that separates them will fail, and we'll always have dependence on $\lambda_1 + \lambda_2$.

3 Basketball

3.1 random variables

Who's Shimshon? I'm assuming this is supposed to be Gal :)

We can define S_1, S_2, S_3 to be the score achieved through penalties, inside, and outside throws. This means the total score he earned is

$$S = S_1 + S_2 + S_3 \quad (18)$$

but the number of successful throws is distributed binomial so

$$S_1 \sim \text{Bin}(N_1, q_1), \quad S_2 \sim \text{Bin}(N_2, q_2), \quad S_3 \sim \text{Bin}(N_3, q_3) \quad (19)$$

noting the large N s are different from small n s. N represents an unknown number of attempts for the distribution, n is the number actually attempted.

3.2 Average score

Who's Shimshon? I'm assuming this is supposed to be Gal :)

$$E[S] = E[S_1 + 2S_2 + 3S_3] = E[S_1] + 2E[S_2] + 3E[S_3] \quad (20)$$

but

$$E[S_1] = n_1 q_1 = 6 \cdot 0.75 = 4.5, \quad E[S_2] = 6, \quad E[S_3] = 4 \quad (21)$$

so

$$E[S] = 28.5 \quad (22)$$

and the variance is, since each set is independent of the others

$$V[S] = V[S_1 + 2S_2 + 3S_3] = V[S_1] + 4V[S_2] + 9V[S_3] = n_1 q_1 (1 - q_1) + 4n_2 q_2 (1 - q_2) + 9n_3 q_3 (1 - q_3) \quad (23)$$

$$= 19.08 \implies \sigma = 4.37 \quad (24)$$

3.3 Most probable throws

The assumption H_0 implies

$$0.75S_1 + 2 \cdot 0.75S_2 + 3 \cdot 0.66S_3 = 24 \quad (25)$$

together with

$$0.25S_1 + 0.25S_2 + 0.34S_3 = 5 \quad (26)$$

and

$$S_1 + S_2 + S_3 = 19 \quad (27)$$

which is three equations for three variables so

$$\begin{pmatrix} 1/2 & 3/2 & 2 \\ 1/4 & 1/4 & 1/3 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} S_1 \\ S_2 \\ S_3 \end{pmatrix} = \begin{pmatrix} 24 \\ 5 \\ 19 \end{pmatrix} \quad (28)$$

throwing into wolfram alpha we get

$$S_1 = 8, \quad S_2 = 8, \quad S_3 = 3 \quad (29)$$

3.4 Extension

Who's Shimshon? I'm assuming this is supposed to be Gal :)

after the extension the numbers are

$$S_1 = 10, \quad S_2 = 8, \quad S_3 = 6 \quad (30)$$

so the p-value is

$$P(S_1 < 8 \cdot 0.75, S_2 < 8 \cdot 0.75, S_3 < 3 \cdot 0.66 | H_0) \quad (31)$$

$$= P(S_1 < 6, S_2 < 6, S_3 < 2 | H_0) \quad (32)$$

$$= P(S_1 < 6)P(S_2 < 6)P(S_3 < 2) \quad (33)$$

which can be directly computed using binomial distribution probability distributions.

4 Signal and Noise

The error on the parameters means

$$b = 8 \pm 0.56, \quad \phi = 0.5 \pm 0.035, \quad \omega = \frac{1}{18} \pm 0.004 \quad (34)$$

4.1 signal and noise

we can find the amount of noise by integrating over the interesting regions

$$\int dx B = b \left(x - \frac{\cos(\omega x + \phi)}{\omega} \right) + C \quad (35)$$

so

$$B_1 = \int_{-14}^{-5} B dx = b \left(9 - \frac{\cos(-5\omega + \phi)}{\omega} + \frac{\cos(-14\omega + \phi)}{\omega} \right) = 70.02 \quad (36)$$

$$B_2 = \int_5^{14} B dx = b \left(9 + \frac{\cos(5\omega + \phi)}{\omega} - \frac{\cos(14\omega + \phi)}{\omega} \right) = 133 \quad (37)$$

and the errors

$$\sigma_{B_1}^2 = \left(\frac{\partial B_1}{\partial b} \right)^2 \sigma_b^2 + \left(\frac{\partial B_1}{\partial \omega} \right)^2 \sigma_\omega^2 + \left(\frac{\partial B_1}{\partial \phi} \right)^2 \sigma_\phi^2 \quad (38)$$

$$= (8.72\sigma_b)^2 + (-676\sigma_\omega)^2 + (71\sigma_\phi)^2 = 37.55 \quad (39)$$

and

$$\sigma_{B_1}^2 = \left(\frac{\partial B_1}{\partial b} \right)^2 \sigma_b^2 + \left(\frac{\partial B_1}{\partial \omega} \right)^2 \sigma_\omega^2 + \left(\frac{\partial B_1}{\partial \phi} \right)^2 \sigma_\phi^2 \quad (40)$$

$$= (16.63\sigma_b)^2 + (327\sigma_\omega)^2 + (37\sigma_\phi)^2 = 90.05 \quad (41)$$

so

$$B_1 = 70.02 \pm 6.13, \quad B_2 = 133 \pm 9.49 \quad (42)$$

so the signal is

$$S_1 = N_1 - B_1 = 84 - 70.02 = 13.98 \pm 6.13 \pm \sqrt{13.98}, \quad S_2 = 147 - 133 = 14 \pm 9.49 \pm \sqrt{14} \quad (43)$$

and

$$seg_1 = \frac{S_1}{\sqrt{B_1}} = 1.67, \quad seg_2 = \frac{S_2}{\sqrt{B_2}} = 1.21 \quad (44)$$

4.2 Joint significance

the total noise under both signals is

$$B = B_1 + B_2 = 203.02 \quad (45)$$

so the total signal is

$$S = N_1 + N_2 - B = 84 + 147 - 203.02 = 27.98 \quad (46)$$

so the joint significance is

$$seg = \frac{S}{\sqrt{B}} = 1.96 \quad (47)$$

4.3 change in significance

we can see that by looking at both signals together, the significance increases.

4.4 recommendation

There are three main improvements that could be made:

Longer Measurement By measuring for a longer time we capture more background allowing us to model it better and improve our method to remove it

More Measurements By repeating the measurements we increase the significance of the signal

More Accurate Measurements Since we know the position of the signal we can focus the accuracy on those regions

4.5 24 hours

Background The background will of course increase proportionally to the increase in time, so 24 times.

Signal We capture the signal 24 times so it also increases by 24 times.

Significance As we saw in a previous part, more measurements of the signal increase its significance, and that is what will happen here too.